



ਆਈ.ਕੇ.ਗੁਜਰਾਲ ਪੰਜਾਬ ਟੈਕਨੀਕਲ ਯੂਨੀਵਰਸਿਟੀ  
I.K. Gujral Punjab Technical University  
Formerly  
Punjab Technical University



Ref. No. 10274391/1

Dated 17 Jun 2025

This is to certify that **Mr./Ms. Harivansh Bhardwaj** S/o/D/o Sh. **Vipin Kumar** has passed **Bachelor of Technology (Computer Science & Engineering)** from **I.K. Gujral Punjab Technical University Campus, Mohali** an affiliated college of this University, under University Roll No. **2127966**. The detail of grade are given as under:-

| FIRST Semester EDP No. 1001688406 Nov-2021           |    |                     |         |
|------------------------------------------------------|----|---------------------|---------|
| Subject Name                                         | TP | Grade               | Credits |
| Mentoring and Professional Development (BMPD-101-18) | P  | S                   | 0       |
| Mathematics-I (BTAM-104-18)                          | T  | O                   | 4       |
| Chemistry-I (BTCH-101-18)                            | T  | B+                  | 4       |
| Chemistry-I(Lab) (BTCH-102-18)                       | P  | B+                  | 1.5     |
| English (BTHU-101-18)                                | T  | A                   | 2       |
| English(Lab) (BTHU-102-18)                           | P  | A+                  | 1       |
| Workshop/Manufacturing Practices (BTMP-101-18)       | P  | A+                  | 3       |
| Programming for Problem solving (BTPS-101-18)        | T  | A                   | 3       |
| Programming for Problem solving(Lab) (BTPS-102-18)   | P  | A                   | 2       |
| Semester Grade Point Average (SGPA) : 8.32           |    | Earned Credits 20.5 |         |

| THIRD Semester EDP No. 1001915342 Nov-2022                                             |    |                   |         |
|----------------------------------------------------------------------------------------|----|-------------------|---------|
| Subject Name                                                                           | TP | Grade             | Credits |
| Mathematics-III (BTAM304-18)                                                           | T  | B                 | 3       |
| Data structure & Algorithms (BTCS301-18)                                               | T  | B+                | 3       |
| Object Oriented Programming (BTCS302-18)                                               | T  | B+                | 3       |
| Data Structure & Algorithms Lab (BTCS303-18)                                           | P  | B+                | 2       |
| Object Oriented Programming lab. (BTCS304-18)                                          | P  | A+                | 2       |
| IT Workshop (BTCS305-18)                                                               | P  | A+                | 1       |
| Digital Electronics (BTES301-18)                                                       | T  | B+                | 3       |
| Digital Electronics Lab (BTES302-18)                                                   | P  | A                 | 1       |
| Foundation Course in Humanities (Development of Societies/Philosophy) (HSMC101/102-18) | T  | A                 | 3       |
| Summer Institutional Training (SIT)                                                    | P  | S                 | 0       |
| Semester Grade Point Average (SGPA) : 7.33                                             |    | Earned Credits 21 |         |

| FIFTH Semester EDP No. 1002247822 Nov-2023                         |    |                   |         |
|--------------------------------------------------------------------|----|-------------------|---------|
| Subject Name                                                       | TP | Grade             | Credits |
| Industrial Training (78336)                                        | P  | O                 | 0       |
| Database Management Systems (BTCS 501-18)                          | T  | A+                | 3       |
| Formal Language & Automata Theory (BTCS 502-18)                    | T  | B                 | 3       |
| Software Engineering (BTCS 503-18)                                 | T  | B+                | 3       |
| Computer Networks (BTCS 504-18)                                    | T  | B+                | 3       |
| Database Management Systems Lab (BTCS 505-18)                      | P  | A+                | 2       |
| Software Engineering Lab (BTCS 506-18)                             | P  | A+                | 1       |
| Computer Networks Lab (BTCS 507-18)                                | P  | A+                | 1       |
| Programming in Python (BTCS 510-18)                                | T  | A                 | 3       |
| Programming in Python Lab (BTCS 513-18)                            | P  | A+                | 1       |
| Enterprise Resource Planning (BTES 501-18)                         | T  | A                 | 3       |
| Constitution of India/Essence of Indian Traditional Knowledge (MC) | P  | O                 | 0       |
| Semester Grade Point Average (SGPA) : 7.83                         |    | Earned Credits 23 |         |

| SEVENTH Semester EDP No. 1002639580 Nov-2024   |    |                   |         |
|------------------------------------------------|----|-------------------|---------|
| Subject Name                                   | TP | Grade             | Credits |
| Project-II (BTCS-703-18)                       | P  | A+                | 6       |
| Network Security and Cryptography (BTCS701-18) | T  | B                 | 3       |
| Data Mining and Data Warehousing (BTCS702-18)  | T  | A+                | 3       |
| Deep Learning (BTCS704-18)                     | T  | B+                | 3       |
| Deep Learning Lab (BTCS705-18)                 | P  | B+                | 1       |
| Blockchain Technologies (BTCS712-18)           | T  | B+                | 3       |
| Blockchain Technologies lab (BTCS713-18)       | P  | A                 | 1       |
| Road Safety (OECE-703-18)                      | T  | A                 | 3       |
| Semester Grade Point Average (SGPA) : 7.83     |    | Earned Credits 23 |         |

Cumulative Grade Point Average CGPA : 7.84

Note:- Medium of Instruction is English

EDP Cell  
Prepared By:  Checked By:

  
AR(Exams)

  
Deputy Controller Exams

  
Controller of Examinations

| SECOND Semester EDP No. 1001798679 April-2022        |    |                     |         |
|------------------------------------------------------|----|---------------------|---------|
| Subject Name                                         | TP | Grade               | Credits |
| Mentoring and Professional Development (BMPD-201-18) | P  | S                   | 0       |
| Mathematics-II (BTAM204-18)                          | T  | B                   | 4       |
| Basic Electrical Engineering (BTEE-101-18)           | T  | B                   | 4       |
| Basic Electrical Engineering(Lab) (BTEE-102-18)      | P  | A+                  | 1       |
| Engineering Graphics & Design (BTME101-21)           | T  | C                   | 3       |
| Semi-Conductor Physics (BTPH-104-18)                 | T  | B                   | 4       |
| Semi-Conductor Physics Lab (BTPH-114-18)             | P  | A                   | 1.5     |
| Semester Grade Point Average (SGPA) : 6.17           |    | Earned Credits 17.5 |         |

| FOURTH Semester EDP No. 1002258348 Nov-2023            |    |                   |         |
|--------------------------------------------------------|----|-------------------|---------|
| Subject Name                                           | TP | Grade             | Credits |
| Discrete Mathematics (BTCS-401-18)                     | T  | A                 | 4       |
| Operating Systems (BTCS-402-18)                        | T  | B+                | 3       |
| Design & Analysis of Algorithms (BTCS-403-18)          | T  | A                 | 3       |
| Operating Systems Lab (BTCS-404-18)                    | P  | A+                | 2       |
| Design & Analysis of Algorithms Lab (BTCS-405-18)      | P  | A+                | 2       |
| Computer Organization & Architecture (BTES-401-18)     | T  | B                 | 3       |
| Computer Organization & Architecture Lab (BTES-402-18) | P  | A+                | 1       |
| Environmental Sciences (EVS 101-18)                    | P  | B+                | 0       |
| Universal Human Values (HSMC-122-18)                   | T  | A+                | 3       |
| Semester Grade Point Average (SGPA) : 7.95             |    | Earned Credits 21 |         |

| SIXTH Semester EDP No. 1002487536 April-2024 |    |                   |         |
|----------------------------------------------|----|-------------------|---------|
| Subject Name                                 | TP | Grade             | Credits |
| Compiler Design (BTCS601-18)                 | T  | A                 | 3       |
| Artificial Intelligence (BTCS602-18)         | T  | A                 | 3       |
| Project-I (BTCS603-18)                       | P  | O                 | 3       |
| Compiler Design Lab (BTCS604-18)             | P  | A+                | 1       |
| Artificial Intelligence Lab (BTCS605-18)     | P  | O                 | 1       |
| Cloud Computing (BTCS612-18)                 | T  | B+                | 3       |
| Cloud Computing Lab (BTCS613-18)             | P  | A+                | 1       |
| Machine Learning (BTCS618-18)                | T  | A                 | 3       |
| Machine Learning Lab (BTCS619-18)            | P  | O                 | 1       |
| Product Design and Development (BTME614-18)  | T  | B+                | 3       |
| Semester Grade Point Average (SGPA) : 8.27   |    | Earned Credits 22 |         |

| EIGHTH Semester EDP No. 1002662565 April-2025 |    |                   |         |
|-----------------------------------------------|----|-------------------|---------|
| Subject Name                                  | TP | Grade             | Credits |
| Semester Training (BTCS801-18)                | P  | A+                | 16      |
| Semester Grade Point Average (SGPA) : 9       |    | Earned Credits 16 |         |